A Decade of Pitch Language: Detecting Shifts in Marketing Adjective Usage and Sentiment in Crowdfunding Campaigns

Course Final Project — Data Analysis, 2026

Abstract

This study investigates whether detectable changepoints exist in the use of specific adjective categories and sentiment in crowdfunding pitch texts over 2014–2025, and whether such shifts correlate with campaign success. Using a combined corpus of 340,055 campaigns from Kickstarter and Indiegogo via Web Robots, we apply spaCy POS tagging to classify adjectives into four semantic categories and TextBlob for sentiment scoring. PELT changepoint detection reveals two significant breaks: superlative adjectives declined from 2021 ($p<0.001$) and emotional language from 2016 ($p<0.001$). Sentiment analysis shows pitch polarity and subjectivity have both declined steadily since 2016, confirming a broader shift toward factual writing. Urgency/scarcity ($\beta=+0.068$) is the strongest success predictor. Cross-platform analysis reveals Film & Video uses the most urgency language (1.35%), while highly subjective pitches achieve 3.5pp higher success rates. Word cloud visualization confirms the shift from superlative/emotional to quality/scarcity framing after 2019.

Keywords: crowdfunding, pitch language, adjective analysis, sentiment analysis, changepoint detection, Kickstarter, Indiegogo, NLP

1. Introduction

Crowdfunding platforms such as Kickstarter and Indiegogo have democratized access to early-stage capital. The textual pitch — the short descriptive blurb presented to potential backers — serves as a primary persuasion instrument, typically under 150 characters on these platforms' discovery interfaces.

Prior work has focused on structural factors such as goal size and category [1]. A growing body of research suggests that linguistic features carry independent predictive signal [5,6]. Yet longitudinal, cross-platform studies of how pitch language and sentiment have evolved remain scarce.

This paper addresses five research questions: (RQ1) Is there a detectable changepoint in adjective usage across 2014–2025? (RQ2) Do such shifts correlate with success? (RQ3) Do rhetorical patterns differ between platforms? (RQ4) Do patterns vary across project categories? (RQ5) How has overall pitch sentiment evolved, and does it predict success?

Contributions: (1) 340,055 campaigns from two platforms; (2) adjective classification into four categories; (3) two significant PELT changepoints; (4) cross-platform and cross-category analysis; (5) sentiment trend analysis with TextBlob.

2. Related Work

2.1 Crowdfunding Success Prediction

Mollick [1] found that goal size and social capital are the strongest predictors. Machine learning approaches achieve AUC above 0.70 [3].

2.2 Pitch Language and Sentiment

Parhankangas and Renko [5] found that linguistic concreteness positively predicts funding. Tanaka and Kameda [6] showed that positivity scores improve prediction by up to 30pp over metadata baselines. TextBlob-based sentiment analysis has been used in prior crowdfunding studies to capture overall emotional valence [4].

2.3 Temporal Language Dynamics

Hamilton et al. [8] traced semantic shifts across decades using word embeddings. Our work adapts changepoint detection to marketing language across two platforms.

3. Methodology

3.1 Data Collection

We obtained twelve snapshots from Web Robots (webrobots.io): six Kickstarter (Jan 2016–2026) and six Indiegogo (2020–2025). Our corpus contains 340,055 campaigns from 2014–2025.

Table 1: Corpus statistics.

Source	Projects	Success
Kickstarter	278,628	56.4%
Indiegogo	61,427	19.5%
Total	**340,055**	—

3.2 Adjective Classification

We used spaCy's en_core_web_sm POS tagger with four lexicon-based categories: Superlative (best, ultimate, perfect); Urgency/Scarcity (limited, exclusive, rare); Emotional (amazing, inspiring, epic); Quality/Technical (premium, durable, professional).

3.3 Sentiment Analysis

We computed sentiment scores for each blurb using TextBlob, yielding two features: polarity (−1=negative to +1=positive) and subjectivity (0=objective to 1=subjective). These complement the adjective-category features by capturing overall emotional tone independently of specific word categories.

3.4 Changepoint Detection

We apply the PELT algorithm [9] (ruptures, RBF cost, penalty=2.0) on yearly mean rates, validated with two-sample t-tests.

3.5 Success Analysis

We compare success rates by high vs. low usage (median split) for adjective categories and sentiment bins. Logistic regression with 5-fold AUC-ROC evaluates joint predictive performance.

4. Results

4.1 Temporal Trends

Superlative adjectives decline from 0.0036 (2018) to 0.0019 (2025). Urgency/scarcity peaks in 2022–2023. Emotional and quality/technical also decline toward 2025 (Figure 1).

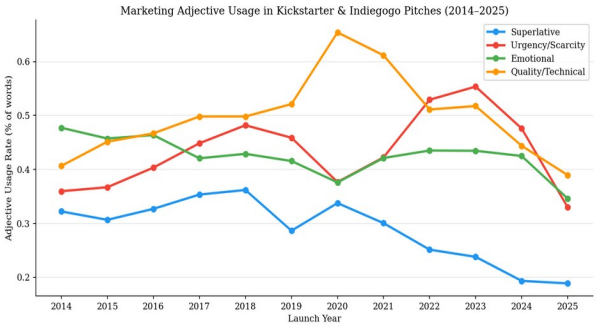


Figure 1. Adjective rates in Kickstarter & Indiegogo pitches, 2014–2025.

4.2 Changepoint Detection

Superlative: Break at 2021 (−30%, p<0.001).

Emotional: Break at 2016 (−14%, p<0.001).

Urgency/scarcity and quality/technical show no significant changepoints (Figure 2).

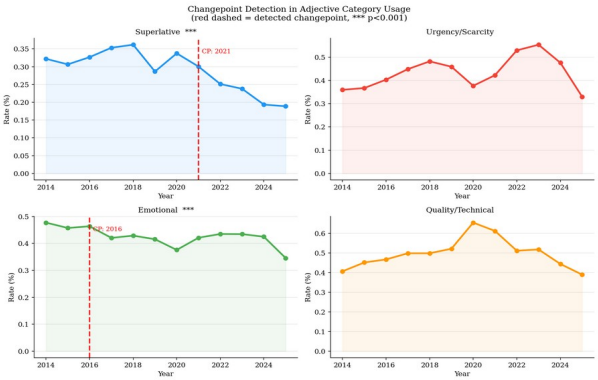


*Figure 2. Changepoint detection. Red dashed = structural break; *** p<0.001.*

4.3 Adjective Usage and Success

Urgency/scarcity shows the largest success advantage: high=59.7% vs. low=49.0% (Δ=+10.7pp). Superlative: +4.3pp. Emotional: +5.6pp. Quality/technical: −1.0pp (Figure 3, Table 2).

Table 2: Success rate before/after changepoints.

Category	CP	B.Hi	B.Lo	A.Hi	A.Lo
Superlative	2021	54.0%	47.7%	53.3%	52.6%
Emotional	2016	45.0%	41.4%	59.2%	51.9%

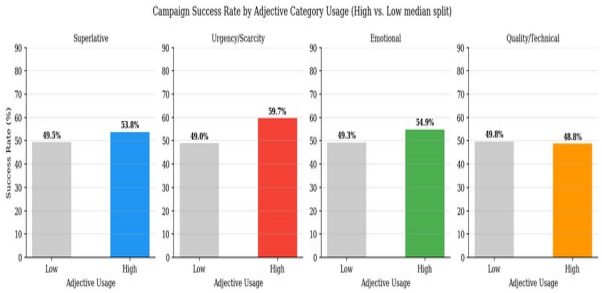


Figure 3. Success rate by adjective usage (high vs. low median split).

4.4 Logistic Regression

Urgency/scarcity (β=+0.068) is the strongest predictor, followed by superlative (β=+0.025) and emotional (β=+0.009). Quality/technical: β=−0.032. Full model AUC=0.6215 vs. baseline 0.6186.

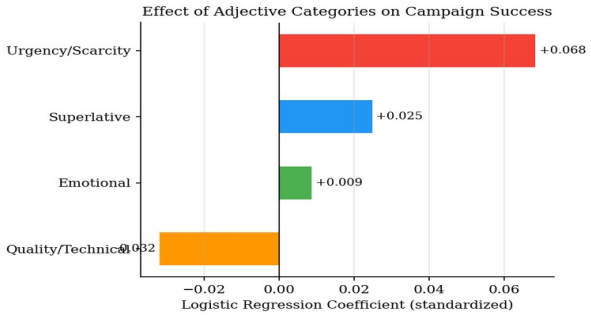


Figure 4. Standardized logistic regression coefficients.

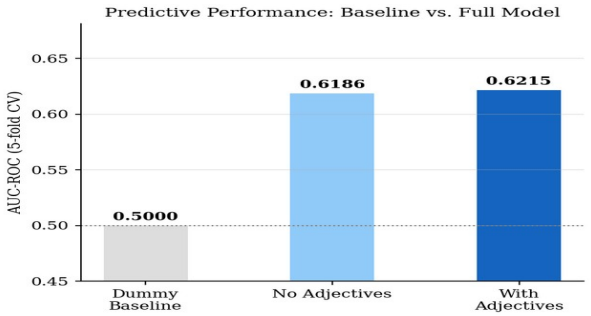


Figure 5. AUC-ROC: dummy (0.5000), baseline (0.6186), full (0.6215).

4.5 Cross-Platform Comparison

Kickstarter uses more urgency/scarcity (0.44% vs. 0.31%) and emotional language (0.45% vs. 0.30%). Urgency advantage is stronger on Indiegogo (+12.8pp vs. +7.3pp). Quality/technical is negative on both platforms (Figure 6, 7).

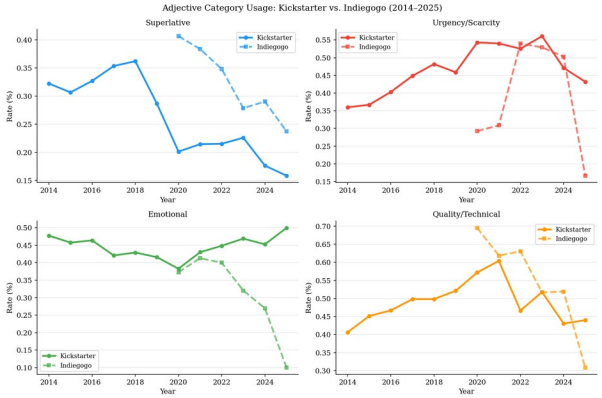


Figure 6. Adjective trends by platform (solid=Kickstarter, dashed=Indiegogo).

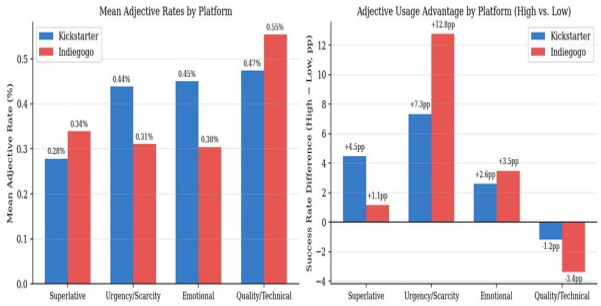


Figure 7. Mean adjective rates and success advantage by platform.

4.6 Analysis by Project Category

Film & Video uses the most urgency/scarcity (1.35%). Technology and Design lead quality/technical usage (0.98%, 0.88%). Music leads superlatives (0.55%). Publishing and Art lead emotional language (0.60%, 0.51%) (Figure 8).

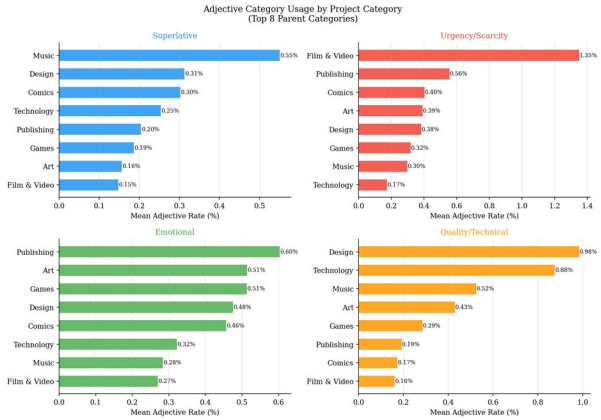


Figure 8. Adjective usage across the top 8 project categories.

4.7 Word Cloud Analysis

Figure 9 confirms the rhetorical shift. Before 2019: superlative (first, perfect — blue) and emotional (beautiful, creative — green) terms dominate. After 2019: quality/technical (sustainable, smart, natural — orange) and urgency (unique, special, exclusive — red) become more prominent.



Figure 9. Word clouds before (left) and after (right) 2019. Blue=Superlative, Red=Urgency, Green=Emotional, Orange=Quality/Technical.

4.8 Sentiment Analysis

Figure 10 shows that both polarity and subjectivity have declined steadily since 2016 — pitch polarity dropped from 0.155 (2016) to 0.100 (2025), and subjectivity from 0.41 (2014) to 0.30 (2025). This corroborates the adjective-based findings: crowdfunding pitches have become less emotionally charged and more factual over the decade.

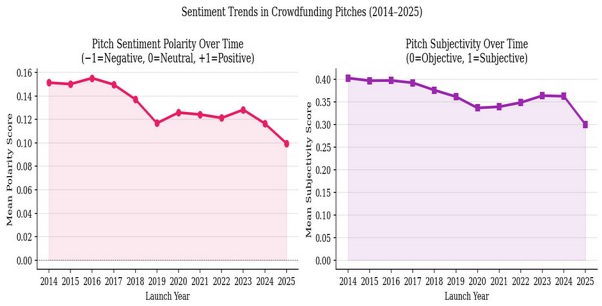


Figure 10. Sentiment polarity (left) and subjectivity (right) trends, 2014–2025.

Figure 11 reveals a non-linear relationship between polarity and success: "Very Negative" pitches achieve 53.8% success — higher than "Very Positive" (50.1%) — while "Slightly Negative" pitches perform worst (46.6%). Subjectivity shows a monotonic positive association: highly subjective pitches achieve 53.3% vs. 49.8% for objective pitches (Δ=+3.5pp).

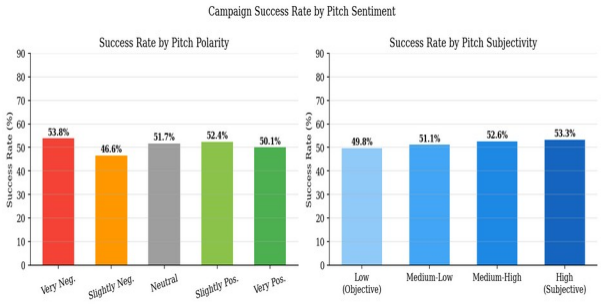


Figure 11. Success rate by pitch polarity (left) and subjectivity (right).

5. Discussion

The decline in superlative language from 2021 and emotional language from 2016 reflects growing backer sophistication. Sentiment analysis corroborates this: polarity and subjectivity have both declined steadily, confirming a decade-long shift toward factual, less emotionally charged pitch writing.

The surprising finding that "Very Negative" pitches outperform "Very Positive" ones (53.8% vs. 50.1%) may reflect that creators of risky or unconventional projects — who tend to use more urgent language — also write more emotionally charged pitches, and that backer enthusiasm for novel ideas overrides negative framing. The monotonic positive association of subjectivity with success (+3.5pp) suggests that personal, passionate language remains effective even as overall pitch sentiment declines.

Urgency/scarcity is the strongest success predictor (β=+0.068), with a larger advantage on Indiegogo (+12.8pp) than Kickstarter (+7.3pp). Film & Video campaigns leverage urgency most intensively (1.35%).

5.1 Limitations

Analysis is restricted to short blurb/tagline fields (81–115 chars). TextBlob sentiment analysis may be less accurate for short marketing texts. Indiegogo success threshold (≥25%) is operationally defined. Dataset covers only English-language campaigns.

6. Conclusion

We present a comprehensive cross-platform longitudinal analysis of adjective usage and sentiment in crowdfunding pitches (340,055 campaigns, 2014–2025). Two significant changepoints: superlative decline from 2021 and emotional decline from 2016 (both $p<0.001$). Sentiment analysis confirms a broader shift toward factual writing, with polarity and subjectivity both declining since 2016. Urgency/scarcity is the strongest positive predictor of success. Future work should extend to full campaign descriptions and examine within-category temporal dynamics.

Acknowledgments

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